

Bayesian Inference, WEEK 01

Lecture Notes

1: Bayesian Inference , Classical Inference Overview

$$\begin{array}{ll} \text{BAYESIAN INFERENCE} & 95\% \text{ CI is } \left(\bar{X} - 1.96 \frac{s}{\sqrt{n}}, \bar{X} + 1.96 \frac{s}{\sqrt{n}} \right) \quad \checkmark \\ & (10, 20) \end{array}$$

Inference: Classical Inference

$$\left\{ \text{ESTIMATION} \left\{ \begin{array}{l} \rightarrow \text{POINT ESTIMATOR} \\ \rightarrow \text{Confidence Interval} \end{array} \right. \right.$$

$$X_n \stackrel{iid}{\sim} f(\cdot | \underline{\theta})$$

$$N(\underline{\mu}, 1) \quad \bar{X}_n \quad \boxed{\text{Var}(T(X))}$$

$$\text{ESTIMATOR} \rightarrow \text{MLE} \quad E(T(X)) \Big\} \rightarrow \text{MSE}$$

$$\boxed{X_1, \dots, X_N} \quad \bar{X} = \frac{\sum X_i}{N} \quad \frac{\sum (X_i - \bar{X})^2}{N}$$

$$\frac{\sum (T^{(i)}(X_n) - \bar{T}_m)^2}{M} \quad \begin{pmatrix} \bar{Y}_1 \\ \vdots \\ \bar{Y}_M \end{pmatrix}$$

$$Y_1, \dots, Y_M$$

$$\bar{X}_n \quad \text{Mean}(Y_1 \dots Y_n) = \frac{\sum X_i}{n}$$

$$\left\{ \begin{array}{l} Y_i \quad T^{(i)}(X_n) \\ \frac{\sum T^{(i)}(X_n)}{M} \\ T^{(1)}(X_n) \\ \vdots \\ T^{(M)}(X_n) = \bar{T}_M \end{array} \right.$$

2: Hypothesis Testing Issues, Practical vs Statistical Significance

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta, 1) \quad \boxed{H_0 : \theta = 0 \text{ vs } H_1 : \theta \neq 0} \quad H_0 : p = 0.5 \text{ vs } H_1 : p \neq 0.5$$

$$\alpha = 0.05 \quad H_0 : \boxed{p = 0.5}$$

$$\boxed{\text{Reject } H_0 \text{ if } \sqrt{n} |\bar{x}| > 1.96}$$

$$\text{Suppose true } \theta = \frac{10^{-10}}{10^{-5}} \quad n = 10^{24} \quad 10^{14} \cdot \sqrt{n} |\bar{x}| \approx 10^{12} \cdot 10^{-10} > 1.96$$

$$0.50000 \quad \boxed{0.49, 0.51} \quad \pi(p \in (0.49, 0.51) | X)$$

$$\underline{\underline{\pi(\theta \in (-10^{-5}, 10^5) | X)}}$$

$$N(\mu, \sigma^2)$$

$$\text{Toss a coin, if head then } \boxed{n = 2}, \text{ if tail then } \boxed{n = 1000} \quad V(\bar{X}) = \frac{1}{2} \left[\frac{\sigma^2}{2} + \frac{\sigma^2}{1000} \right] \approx \boxed{\frac{\sigma^2}{4}}$$

$$\pi(\theta | \underline{\text{Data}}) \quad (\theta - \frac{1}{2}, \theta + \frac{1}{2}) \quad \boxed{X_1 = 2, X_2 = 2}$$

$$U(\theta - \frac{1}{2}, \theta + \frac{1}{2}) \quad \boxed{X_2 - X_1} \quad \checkmark$$

3: Bayes Theorem , Disease Diagnostic Example

Bayes theorem: Suppose a disease affects 2% of a population.

A diagnostic test is available with the following properties:

- If a person has the disease, the test is positive with 0.95 prob.
- If a person doesn't have the disease, the test is positive with 0.10 prob. [false positive]

Let D = Event that a person has the disease.

T^+ : the test result is positive.

$$\text{What is } \boxed{P(D | T^+)} = \frac{P(D \cap T^+)}{P(T^+)} = \frac{P(T^+ | D)P(D)}{P(T^+)} = 0.1624$$

Right column:

$$P(D) = 0.02 \quad P(D^c) = 0.98 \quad [\text{Prior probability}]$$

$$P(T^+ | D) = 0.95, \quad P(T^+ | D^c) = 0.10 \quad [\text{Conditional probabilities}]$$

$$P(T^+) = P(D)P(T^+ | D) + P(D^c)P(T^+ | D^c) = 0.117 \quad [\text{Marginal prob.}]$$

4: Three-Prisoners Problem and Course Evaluation

Three-prisoners problem: A | B | C

2 → A asks the guard whether A will be executed?

$$P(c | b) = \frac{2}{3}$$

$b \leftarrow B$ will be executed

$$P(\text{Pardon for } A) = \frac{1}{3}$$

$$P(\text{Pardon for } A | b) = (a)\frac{1}{3} (b)\frac{1}{2} (c)\frac{2}{3} (d)1 \quad P(\text{Pardon for } A | b) = \frac{P(A, b)}{P(b)}$$

$$P(A, b) = P(A) \times P(b | A)$$

$$P(b) = P(A)P(b | A) + P(B)P(b | B) + P(C)P(b | C) =$$

$$= \frac{1}{3} \times \boxed{\frac{1}{2}} + \frac{1}{3} \times \boxed{0} + \frac{1}{3} \times \boxed{1}$$

$$P(A | b) = \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{3} \times \frac{3}{2}} = \frac{1}{3}$$

Course structure:

25% MID SEM.

50% END SEM

→ Assignment: Read and apply application paper

25% Project / Assignment / quiz Bayesian Data Analysis

(2) $\boxed{3-4}$ (1-2) (Gelman)

(Stan) rstan, pystan

5: Prior, Likelihood, Posterior Revisited

$$P(D) = 0.02 \text{ (prior)}$$

$$P(T^+ | D) = 0.95 \text{ (conditional / likelihood)}$$

$$P(D | T^+) = 0.1624 \text{ (Posterior)}$$

$$P(D) = 0.20 \quad P(D^c) = 0.80 \quad P(D | T^+) = 0.704$$

$$P(D | T^+) = \frac{\boxed{P(D)} P(T^+ | D)}{\boxed{P(T^+)}} \quad P(D^c | T^+) = \frac{P(D^c) P(T^+ | D^c)}{P(T^+)}$$

$$P(D | T^+) \propto \underline{P(D)} P(T^+ | D) \quad P(D^c | T^+) = \frac{\boxed{P(D^c)} \boxed{P(T^+ | D^c)}}{k}$$

$$P(D | T^+) + P(D^c | T^+) = 1 \quad P(D | T^+) = \frac{0.19}{k}$$

$$\boxed{P(D | T^+) \propto P(D) P(T^+ | D)} \quad P(D^c | T^+) = \frac{0.08}{k}$$

$$\boxed{\text{Posterior} \propto \underline{\text{prior}} \times \text{likelihood}}$$

$$H_0 : p = 0.95 \text{ vs. } H_1 : p = 0.10 \quad P(T^+ | D) = 0.95 \quad P(T^+ | D^c) = 0.10$$

Prior (Subjective belief)

$$\boxed{\text{Prior}} \leftarrow \begin{array}{|l|l|} \hline P(H_0) = 0.02 & P(H_1) = 0.98 \\ P(H_0) = 0.20 & P(H_1) = 0.80 \\ P(H_0) = 0.50 & P(H_1) = 0.50 \\ \hline \end{array} \quad \boxed{P(H_0 | T^+) = 0.16 = 0.70}$$

$$\underline{\underline{\text{Cancer}}} \quad \underline{\underline{0.16}} \times \boxed{(C)}$$

6: Paradoxes in Classical Paradigm , Uniform Example

$$\underline{\underline{\text{Cancer}}} \quad \underline{\underline{0.16}} \times \boxed{(C)} \quad 0.16 \times C$$

$$\underline{\text{Paradoxes in classical paradigm}} \quad X \sim U(\theta - \tfrac{1}{2}, \theta + \tfrac{1}{2})$$

$$\begin{array}{|c|} \hline [T_l(X), T_u(X)] \\ \hline \theta \in [\bar{X} - \dots, \bar{X} + \dots] \rightarrow \boxed{\begin{array}{|c|} \hline 95\% \\ 50\% \\ \hline \end{array}} \\ \hline \theta \in [1.5 - c, 1.5 + c] \text{ with } 75\% \text{ CI} \\ \hline \end{array}$$

$$X_1, X_2 \text{ iid}$$

$$\text{let } y_1 = \min\{X_1, X_2\}$$

$$y_2 = \max\{X_1, X_2\}$$

$$\begin{aligned} \forall \theta, P(y_1 < \theta < y_2 | \theta) &= P(x_1 < \theta < x_2) + P(x_2 < \theta < x_1) \\ &= \tfrac{1}{2} \times \tfrac{1}{2} + \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{2}. \end{aligned}$$

$$\therefore (y_1, y_2) \text{ gives you a } 50\% \text{ CI of } \theta.$$

$$\text{Let } x_1 = 7.4, x_2 = 8.0 \quad (7.4, 8.0) \quad 50\% \text{ CI of } \theta.$$

$$\theta - \tfrac{1}{2} < x_1, x_2 < \theta + \tfrac{1}{2} \quad \text{We must have } \theta + \tfrac{1}{2} > 8.0 \Rightarrow \theta > 7.5$$

$$\therefore \text{With certainty, } \theta \in (7.5, 7.9) \quad \theta - \tfrac{1}{2} < 7.4 \Rightarrow \theta < 7.9$$

7: Binomial vs Negative Binomial , Likelihood Principle

12 independent coin tosses are done and you get 9 heads.

$H_0 : p = \frac{1}{2}$ vs $H_1 : p > \frac{1}{2}$

Binomial: $L_1(p) = \binom{12}{9} p^9 (1-p)^3$

$\rightarrow$ p-value = $P(X \geq 9) = 0.075$

$\alpha = 0.05$

Negative Binomial:

$L_2(p) = \binom{11}{9} p^9 (1-p)^3$

$P(X \geq 9) = 0.0325$

Likelihood principle (LP): In making inferences or decision about θ , after x is observed, all relevant information is contained in the likelihood function of the observed x .

Advantages of Bayesian paradigm.

- (1) Prior info can be incorporated
- (2) Inferences conditional on actual data. $P(\theta)P(D_n | \theta)$
- (3) LP
- (4) Posterior $\rightarrow$ No need for separate hypothesis etc.
- (5) Does not rely on asymptotic theory.
- (6) However complex the model is, arbitrarily accurate inference by computational methods.